ENTHUSE, LEADER & ACHIEVER COURSE

PHASE : ALL ENTHUSE, LEADER, ACHIEVER COURSE SRG MAJOR & LAKSHYA PHASE

TARGET : PRE-MEDICAL 2026

Test Type : **SRG-MAJOR** Test Pattern : **NEET (UG)**

TEST DATE : 29-05-2026

ANSWER KEY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	4	3	4	4	3	4	4	4	2	4	4	4	1	1	3	3	3	2	4	4	4	2	2	2	4	1	1	3	1	2
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	4	2	1	3	3	2	4	3	1	4	1	3	2	4	3	3	3	4	4	4	3	2	3	1	4	1	3	1	1	3
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A.	4	3	3	2	2	1	3	4	1	4	2	4	3	4	2	2	3	2	1	3	1	4	2	3	2	4	3	2	1	3
Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	2	1	3	3	2	4	3	4	4	2	2	4	4	2	4	2	3	2	3	3	1	3	4	2	2	1	2	2	2	2
Q.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	3	2	4	4	1	2	1	2	4	2	3	2	4	4	3	3	4	3	2	1	4	4	4	4	1	4	4	1	3	1
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
A.	2	3	1	3	1	4	1	1	2	2	3	2	4	2	1	2	3	2	1	1	4	3	2	1	1	4	3	2	1	4

HINT – SHEET

1. Ans (4)

Three identical rods (same length L and area A).

Thermal conductivity of C = K.

Thermal conductivity of B = 3K (thrice of C).

Thermal conductivity of A = 6K (since B is half of A).

Solution:

Formula: $\frac{L_{total}}{K_{eff}A} = \frac{L_A}{K_AA} + \frac{L_B}{K_BA} + \frac{L_C}{K_CA}$

Calculation: Total length = 3L.

$$\frac{3L}{K_{eff}} = \frac{L}{6K} + \frac{L}{3K} + \frac{L}{K}$$

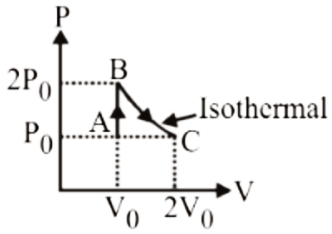
$$\frac{3}{K_{eff}} = \frac{1}{6K} + \frac{2}{6K} + \frac{6}{6K}$$

$$\frac{3}{K_{eff}} = \frac{9}{6K} \Rightarrow \frac{3}{K_{eff}} = \frac{3}{2K}$$

$K_{eff} = 2K$

2. Ans (3)

Solution:



From the diagram (described):

Process A⟶B: Isochoric (volume constant = V_0)

At A: P = P_0, V = V_0

At B: P = $2P_0$, V = V_0

Process B⟶C: Isothermal at temperature corresponding to P = $2P_0$, V = V_0, expanding to V = $2V_0$.

Step 1: A⟶B (Isochoric)

Work done $W_{AB}=0$

Change in internal energy:

$\Delta U_{AB}=nC_v\Delta T$

For diatomic gas, $C_v=\frac{5}{2}R$

From A to B:

$\frac{P_AV_A}{T_A}=\frac{P_BV_B}{T_B}\Rightarrow\frac{P_0V}{T_A}=\frac{2P_0V_0}{T_B}\Rightarrow T_B=2T_A$

SO $\Delta T=T_B-T_A=T_A$

But $nRT_A=P_0V_0\Rightarrow T_A=\frac{2P_0V_0}{nR}$

Thus

$\Delta U_{AB}=n\cdot\frac{5}{2}R\cdot T_A=\frac{5}{2}nR\cdot\frac{P_0V_0}{nR}=\frac{5}{2}P_0V_0$

Heat in A⟶B:

$Q_{AB}=\Delta U_{AB}+W_{AB}=\frac{5}{2}P_0V_0$

Step 2: B⟶C (Isothermal)

For isothermal, $\Delta U_{BC}=0$

Work done $W_{BC}=nRT_B\,In\left(\frac{V_C}{V_B}\right)$

$nRT_B=P_BV_B=(2P_0)(V_0)=2P_0V_0$

$V_C=2V_0,\ V_B=V_0\Rightarrow\frac{V_C}{V_B}=2$

$W_{BC}=2P_0V_0\ In\ 2$

Heat in B⟶C:

$Q_{BC}=\Delta U_{BC}+W_{BC}=0+2P_0V_0\ In\ 2$

Step 3: Total heat

$Q_{total}=QAB+QBC=\frac{5}{2}P_0V_0+2P_0V_0\ In\ 2$

3. Ans (4)

given: $PT=C$, $PV=nRT$

$T\propto PV$

$P(PV)=C$

$P^2V=C$

$PV^{1/2}=C$

$\therefore x=\frac{1}{2}$

$C=C_V+\frac{R}{1-x}$

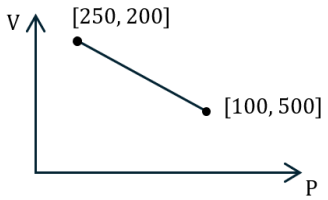
$4.5\,R=\frac{f}{2}R+\frac{R}{1-\frac{1}{2}}$

$2.5\,R=\frac{fR}{2}$

$f=5$

4. Ans (4)

Solution :



$P_1=200$ (KPa) , $V_1=250$ CC

$P_2=500$ KPa, $V_2=100$ CC

$\Delta U=\frac{f}{2}[P_2V_2-P_1V_1]$

$\Delta U=\frac{f}{2}[500\times100-250\times200]$

$\Delta U=0$

5. Ans (3)

Solution :

As we know that, from Newton's law of cooling.

Rate of cooling $\propto(T_{body}-T_{surrounding})$

$\Rightarrow$ That means the assertion is true.

But, this law is only valid when the difference between body temperature & surrounding temperature is not more than 35°C. Therefore reason is not correct.

6. Ans (4)

$Q=mL$

Extra heat is absorbed by Ice to melt from Juice.

That's why Juice will become more chilled.

$(Q=mL_f)$

7. Ans (4)

$\Delta\ell_1=\Delta\ell_2$

$\ell_1\,\alpha_1\,\Delta T=\ell_2\,\alpha_2\,\Delta T$

$\ell_1\alpha_1=\ell_2\alpha_2$

8. Ans (4)

Concept: According to Charles's Law, a direct proportionality between Volume and Temperature ($V\propto T$) indicates an isobaric process where pressure remains constant.

Solution:

From graph

$T=(\tan 45°)\,V$

$T=V$ ...(i)

using ideal gas equation

$PV=\mu RT$ ....(ii)

from (i) & (ii)

$PV=\mu RV$

$P=\mu R=$ constant.

9. Ans (2)

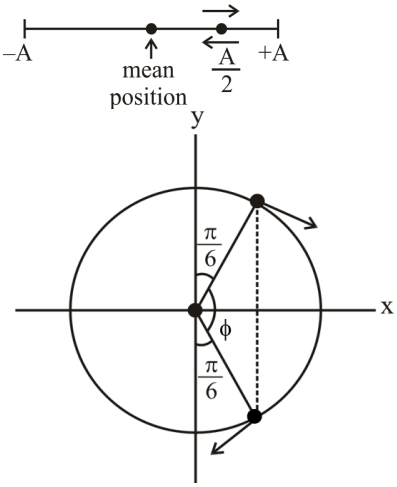
$a_{max.} = g = \omega^2 A \quad (A = d)$

$\Rightarrow \quad \omega^2 = \frac{g}{d}$

$\Rightarrow \quad \omega = \sqrt{\frac{g}{d}}$

or $\quad f = \frac{1}{2\pi}\sqrt{\frac{g}{d}}$

10. Ans (4)



∴ Phase difference between the two particles is

$\phi = \pi - 2 \times \left(\frac{\pi}{6}\right)$

$\phi = \pi - \frac{\pi}{3} = \frac{2\pi}{3}$

$\phi = \frac{2\pi}{3} = \frac{2 \times 180^\circ}{3} = 120^\circ$

11. Ans (4)

Given that: $\ell_1 = \eta\ell_2$

Total length: $\ell = \ell_1 + \ell_2 \quad \ell = \eta\ell_2 + \ell_2$

$\ell = (\eta + 1)\ell_2$

$\ell_2 = \frac{\ell}{\eta + 1}$

$\ell_1 = \frac{\eta\ell}{\eta + 1}$

For a uniform spring: $k\ell$ = constant

$k\ell = k_1\ell_1 = k_2\ell_2$

$k\ell = k_1\left(\frac{\eta\ell}{\eta + 1}\right) = k_2\left(\frac{\ell}{\eta + 1}\right)$

$k_1 = k\frac{\eta + 1}{\eta}$

$k_2 = k(\eta + 1)$

12. Ans (4)

Acceleration is given by $a = \frac{dv}{dt}$. Since velocity (v) is given as a function of position (x), we must use the chain rule:

$a = \frac{dv}{dx} \cdot \frac{dx}{dt}$

Since $v = \frac{dx}{dt}$, the formula becomes $a = v \cdot \frac{dv}{dx}$.

First, let's find $\frac{dv}{dx}$ from v=2(xsinx + cosx):

$\frac{dv}{dx} = 2\left(\frac{d}{dx}(x\sin x) + \frac{d}{dx}(\cos x)\right)$

$\frac{dv}{dx} = 2(1.\sin x + x.\cos x - \sin x) = 2x\cos x$

Now, substitute back into the acceleration formula:

$a = [2(x\sin x + \cos x)] \cdot [2x\cos x]$

We need to find the acceleration at $x=\pi/2$. We know that $\cos(\pi/2)=0$. Since the entire expression for acceleration is multiplied by cosx, the acceleration at $x=\pi/2$ will be zero.

13. Ans (1)

Intensity $I = \frac{P}{A}$

Solution :

Plane wave: Area (A) is constant, so Intensity $\left(I = \frac{P}{A}\right)$ is constant.

Spherical wave: Area ($A = 4\pi r^2$), so Intensity $I = \frac{P}{4\pi r^2}$ is proportional to $\frac{1}{r^2}$

Total power of spherical wave over a sphere of radius r is $I \times A = P$, which is constant.

14. Ans (1)

$\omega_1 = 500\pi \Rightarrow f_1 = \omega_1 / 2\pi = 250$ Hz

$\omega_2 = 506\pi \Rightarrow f_2 = \omega_2 / 2\pi = 253$ Hz

Beats per second

$= |f_2 - f_1| = 3$ beats/s

Amplitudes : $A_1 = 4$, $A_2 = 2$

Maximum amplitude

$A_{max} = A_1 + A_2 = 6$

Minimum amplitude

$A_{min} = |A_1 - A_2| = 2$

Intensity $\propto$ (amplitude)^2

Intensity ratio

$I_{max} : I_{min} = (6^2) : (2^2) = 36 : 4 = 9 : 1$

15. Ans (3)

Let the frequency of the tuning fork be **'f'**. The frequency of a sonometer wire (f_{wire}) is inversely proportional to its length (L), so $f_{wire} \cdot L$ is constant. The shorter wire (20 cm) has a higher frequency, f1, and the longer wire (21 cm) has a lower frequency, f_2. Since the beat frequency (5 Hz) is the same in both cases, the fork's frequency **'f'** must lie between the two wire frequencies.

Therefore, $f_1=f+5$ and $f_2=f-5$.

Using the relation $f_1L_1=f_2L_2$:

$(f+5)\times20=(f-5)\times21$

$20f+100=21f-105$

$205=f$

$\Rightarrow f = 205$ Hz

16. Ans (3)

$80 = \frac{1}{2} \times 10 \times t^2 \Rightarrow t = 4$ sec

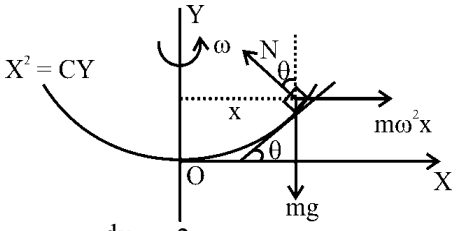
$10 \times 4 + v \times 4 = 60 \Rightarrow v = 5$ m/s

17. Ans (3)

$t^2 = \frac{x-a}{b} \Rightarrow x = a + bt^2$

$\Rightarrow$ constant accelerated motion

18. Ans (2)



$\tan\theta = \frac{dy}{dx} = \frac{2x}{C}$

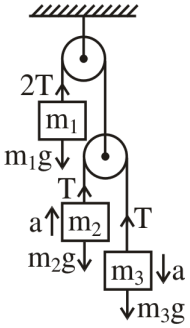
$N\cos\theta = mg$ ....(i)

$N\sin\theta = m\omega^2 x$ ....(ii)

(ii)/(i) , $\tan\theta = \frac{\omega^2 x}{g}$

$\Rightarrow \omega = \sqrt{\frac{2g}{C}}$

19. Ans (4)



$T = \frac{2m_2m_3g}{m_2+m_3}$ ....(i)

$2T = m_1 g$ ....(ii)

$m_1 = \frac{4m_2m_3}{m_2+m_3}$

$\frac{4}{m_1} = \frac{m_2+m_3}{m_2m_3}$

$\frac{4}{m_1} = \frac{1}{m_2} + \frac{1}{m_3}$

20. Ans (4)

$V_1 = u_1 - gt$

$V_2 = u_2 - gt$

Now $T_{f_1} < T_{f_2}$ (T_f is time of flight)

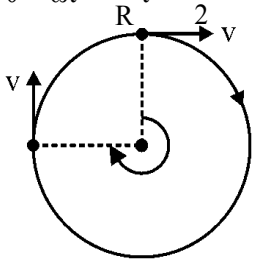
So, till T_{f_1}

$V_2 - V_1 = (u_2 - u_1)$ = constant then after 1 will be on ground and 2 will falling down.

21. Ans (4)

The angular displacement of the particle in t = 1 sec is

$\theta = \omega t = \frac{v}{R}t = \frac{3\pi}{2}$



$\therefore$ The magnitude of impulse by centripetal force in t = 1 second is

= change in momentum = $\sqrt{2}$ mv = $3\sqrt{2}\pi$ Ns.

22. Ans (2)

For marble A, its speed increases more, so it will reach earlier.

23. Ans (2)

After 1st collision, speed = (2 × 5) + 3 = 13 m/s

After 2nd collision, speed = (2 × 3) + 13 = 19 m/s

Final speed = 19 m/s

24. Ans (2)

Direction of friction

=– (Direction of relative velocity of block w.r.t plank)

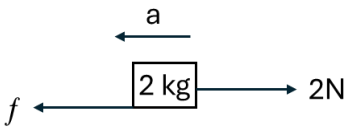
$\vec{V}_{bp} = 3\hat{i} + 4\hat{j}$

$\left(-\hat{V}_{bp}\right) = \dfrac{-\left(3\hat{i} + 4\hat{j}\right)}{5}$

25. Ans (4)

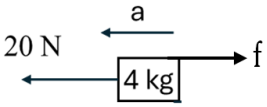
Net force by surface on block will be opposite gravity.

26. Ans (1)



$f - 2 = 2a$ ...(1)

$a = \dfrac{f-2}{2}$



$20 - f = 4a$ ...(2)

Put a in (2)

$20 - f = 4\left(\dfrac{f-2}{2}\right)$

$20 - f = 2f - 4$

$24 = 3f$

$f = 8N$

27. Ans (1)

The last completed reading on the main scale is **2 mm**.

The coinciding thimble division is **31**.

Total reading = Main scale reading + (Thimble division × least count)

Total reading = 2.00 mm + (31 × 0.01 mm) = 2.31 mm

Final answer : Option (1)

28. Ans (3)

$Y = 1.9 \times 10^{11}\ N/m^2$

$Y = 1.9 \times 10^{11} = 1.9 \times 10^{11}\ kg\ m^{-1}\ s^{-2}$

$= 1.9 \times 10^{12}\ gm\ cm^{-1}\ s^{-2}$

29. Ans (1)

The problem asks to determine what 'y' represents if $y = \dfrac{\sqrt{Fx - mv^2}}{\sqrt{m}}$ is given.

$[y] = \dfrac{[mv^2]^{1/2}}{[m]^{1/2}}$

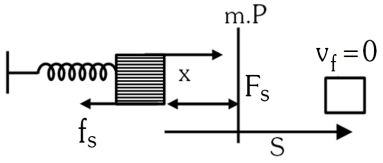
$[y] = [v] = [LT^{-1}]$

30. Ans (2)

For stable $\dfrac{dF}{dx} < 0$

For unstable $\dfrac{dF}{dx} > 0$

31. Ans (4)



By W.E.T.

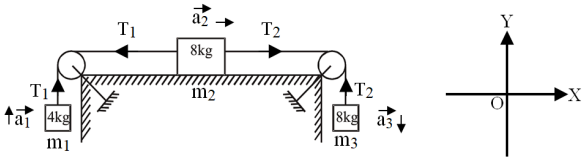
$W_s + W_f = \Delta KE = 0$

$\dfrac{1}{2}Kx^2 - \mu mgS = 0$

$\dfrac{1}{2}Kx^2 = \mu mgS$

$S = \dfrac{1}{2}\dfrac{Kx^2}{\mu mg}$

32. Ans (2)



$|\vec{a}_1| = |\vec{a}_2| = |\vec{a}_3| = |a|$

$T_1 - 4g = 4a$ _____(i)

$T_2 - T_1 = 8a$ _____(ii)

$8g - T_2 = 8a$ _____(iii)

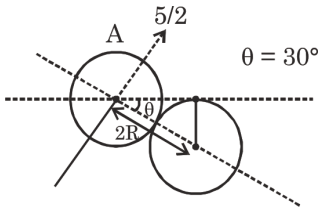
Solving (i), (ii) and (iii) ; we get $|a| = 2\ m/s^2$

$\vec{a}_{cm} = \dfrac{m_1\vec{a}_1 + m_2\vec{a}_2 + m_3\vec{a}_3}{m_1 + m_2 + m_3}$

$= \dfrac{-8\hat{j} + 16\hat{i}}{20}$

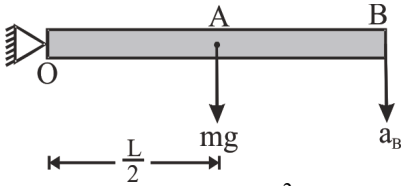
$\Rightarrow \vec{a}_{cm} = -0.4\hat{j} + 0.8\hat{i}$

33. Ans (1)



Velocity perpendicular to impact will not change

34. Ans (3)



$\tau_0 = Mg\frac{L}{2} = I_0\alpha = \frac{ML^2}{3}\alpha$

$\alpha = \frac{3g}{2L}$

$a_B = L\alpha = \frac{3g}{2}$

35. Ans (3)

Initial K.E. $= \frac{1}{2}mu^2 + \frac{1}{2}I\omega^2$

$KEi = \frac{1}{2}mu^2 + \frac{1}{2}\left(\frac{2}{5}mr^2\right)\omega^2, (u = \omega r)$

$KEi = \frac{1}{2}mu^2 + \frac{1}{5}mu^2 = \frac{7}{10}mu^2$

Total KE converts to P.E. at maximum height (h)

$\Rightarrow mgh = \frac{7}{10}mu^2$

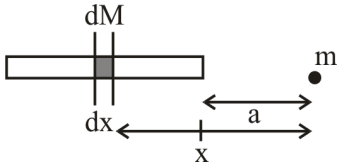
$h = \frac{7u^2}{10g}$

36. Ans (2)

For triangular lamina

$I = \frac{Mh^2}{6}$ here $h = \frac{\ell}{\sqrt{2}}$

37. Ans (4)



$dM = \frac{M}{L}dx$

$F = \int_a^{a+L} \frac{GMm\ dx}{x^2.L}$

$F = \frac{GMm}{L}\int_a^{a+L}\frac{dx}{x^2} = \frac{GMm}{a(a+L)}$

38. Ans (3)

Assertion (A) is true as a uniform shell acts as a point mass for external points.

The perpendicular components of gravitational force from various regions of the shell cancel out due to symmetry.

Only the component along the line joining the point mass to the center remains.

Reason (R) is false because it implies non-vanishing perpendicular components.

39. Ans (1)

$T.E. = -K.E. = \frac{P.E.}{2}$

40. Ans (4)

$U = \frac{1}{2} \times \text{stress} \times \text{strain}$

$U = \frac{1}{2}Y(\text{strain})^2$

$U = \frac{1}{2}Y\left(\frac{\Delta\ell}{L}\right)^2$

$U \propto \left(\frac{\Delta\ell}{L}\right)^2$

$\frac{U_1}{U_2} = \frac{\left(\frac{\ell}{L}\right)^2}{\left(\frac{2\ell}{2L}\right)^2} = 1$

41. Ans (1)

Diameter of big drop = D (given)

Diameter of each small drop = d (assume)

Volume conservation:

$\frac{4}{3}\pi\left(\frac{D}{2}\right)^3 = 27 \times \frac{4}{3}\pi\left(\frac{d}{2}\right)^3$

$d = \frac{D}{3}$

Change in energy $\Delta E = T\,\Delta A$

$\Delta E = T[A_{final} - A_{initial}]$

$\Delta E = T[n \times 4\pi r^2 - 4\pi R^2]$

$\Delta E = T\left[27 \times 4\pi\left(\frac{d}{2}\right)^2 - 4\pi\left(\frac{D}{2}\right)^2\right]$

$\Delta E = T\left[27 \times \pi d^2 - \pi D^2\right]$

$\Delta E = \pi T\left[27\left(\frac{D}{3}\right)^2 - D^2\right]$

$\Delta E = \pi T\left[3D^2 - D^2\right] = 2\pi TD^2$

42. Ans (3)

From continuity:

$A_1V_1 = A_2V_2 \Rightarrow 2 \times 10^{-2} \times 2 = 0.01 \times V_2$

$V_2 = 4$ m/s

Applying Bernoulli's theorem (horizontal pipe):

$P_1 + \frac{1}{2}\rho V_1^2 = P_2 + \frac{1}{2}\rho V_2^2$

Take $\rho = 1000$ kg/m^3:

$4 \times 10^4 + \frac{1}{2}(1000)(2^2) = P_2 + \frac{1}{2}(1000)(4^2)$

$40000 + 2000 = P_2 + 8000$

$P_2 = 42000 - 8000 = 34000$ Pa

43. Ans (2)

Solution:

For topping $\tau_F \geqslant \tau_{mg}$

$F\left(\frac{3}{4}\right) \geqslant 30\left(\frac{1}{2}\right)$

$F \geq 20$ N

$F_{min} = 20$ N

44. Ans (4)

Solution:

At latitude λ, effective outward acceleration acting radially is:

Centrifugal acceleration = $\omega^2 R \cos\lambda$

Radial (vertical) component = $\omega^2 R \cos^2\lambda$

For weightlessness,

$g = \omega^2 R \cos^2\lambda$

So, $\omega = \sqrt{(g / (R\cos^2\lambda))}$

For $\lambda = 30°$

$\cos 30° = \sqrt{3}/2$

$\cos^2 30° = 3/4$

Therefore,

$\omega = \sqrt{(g / (R \times 3/4))}$

$\omega = \sqrt{(4g / 3R)}$

Final Answer: Option (4)

45. Ans (3)

Terminal speed $V_t = \frac{2}{9}\frac{(\rho_S - \rho_W)r^2}{\eta}g$ where ρ and σ are the densities of spherical body and medium respectively, r is the radius of the spherical body and η is the coefficient of viscosity of the medium.

Here r = 1 mm = 10^{-3}

Specific gravity of Aℓ = 2.7

Density of Al, $\rho = 2.7 \times 10^3$ kg/m^3

Density of water $\sigma = 10^3$ kg/m^3

$\eta = 8 \times 10^{-4}$, g = 9.8 m/s^2

$\therefore V_r = \frac{2(10^{-3})^2 \times (2.7 \times 10^3 - 10^3) \times 9.8}{9 \times 8 \times 10^{-4}}$

$V_t = 4.6$ m/s

46. Ans (3)

Explain Question:

This question asking about hypervalent compounds exceed the octet.

Concept:

Hypervalent compounds

Solution:

PCl_5, SF_6, ClF_3 ⟶ more than 8 electrons

BF_3 ⟶ less than 8 electrons ⟶ not hypervalent

Final answer: Option (3)

47. Ans (3)

Explanation

- Analyze the structure and magnetic properties of the ions present in each solid compound.

Concept

1. $PCl_5(s)$: Exists as $[PCl_4]^+$ (tetrahedral) and $[PCl_6]^-$ (octahedral). The statement is correct.
2. $N_2O_5(s)$: Exists as $[NO_2]^+$ (linear) and $[NO_3]^-$ (planar trigonal). The statement is correct.
3. $I_2(s)$: Solid iodine (I_2) is a molecular solid, not an ionic solid, because it consists of I_2 molecules held together by weak van der Waals forces, not by ionic bonds.

Both ions are linear, the statement is incorrect.

$Cl_2O_6(s)$: Exists as $[O_2ClO^+][ClO_4]^-$. Both the perchlorate ($[ClO_4]^-$) and the dichlorine dioxide cation ($[O_2ClO]^+$) have all paired electrons, making them diamagnetic. The statement is correct.

48. Ans (4)

Concept:

VSEPR Theory

Solution:

• The molecule MX_3 has 3 bond pairs of electrons. The shape depends on the number of lone pairs on the central atom (M):

• 0 lone pairs: The electron pair geometry is trigonal planar, and the molecular shape is also triangular (or trigonal planar). Example: BF_3.

• 1 lone pair: The electron pair geometry is tetrahedral, and the molecular shape is pyramidal (specifically, trigonal pyramidal). Example: NH_3 (although this is AX_3E type). For MX_3 with one lone pair, consider a hypothetical molecule.

• 2 lone pairs: The electron pair geometry is trigonal bipyramidal, and the molecular shape is T-shape. Example: ClF_3 (AX_3E_2 type).

Final Answer: Option (4)

49. Ans (4)

M.P. of Ionic compound > M.P. of covalent compound.

50. Ans (4)

Question explanation : We need to identify which interaction represents the strongest attraction between molecules or ions.

Concept :

Ion−dipole interactions are stronger than hydrogen bonding and all types of van der Waals forces.

Solution

H_2—H_2: Nonpolar molecules ⟶ weak London dispersion forces.

H−Cl—C_6H_6: Polar + nonpolar ⟶ dipole−induced dipole ⟶ weak.

H−Cl—H−Cl: Dipole−dipole interaction ⟶ moderate strength.

Na^+—Cl−H: Ion−dipole interaction ⟶ strongest among all listed.

Final answer Option 4 (Na^+ — Cl−H)

51. Ans (3)

Concept

52. Ans (2)

MOT

B.O. ∝ B.E.

53. Ans (3)

Explain Question: Match element types with their outer electronic configurations.

Concept: Electronic configuration

Solution:(A) Inert gas ⟶ (Q) $1s^2$ & ns^2np^6

(B) Representative elements ⟶ (P)$ns^{1-2}\, np^{0-5}$

(C) Transition elements ⟶ (S) $(n-1)d^{1-10}\, ns^{0-2}$

(D) Inner transition elements ⟶ (R)$(n-2)\, f^{1-14}$ $(n-1)d^{0-1}ns^2$

Final Answer: Option (3)

54. Ans (1)

F do not have the tendency to donate e^-

55. Ans (4)

IP C < O < N < O^+

56. Ans (1)

(i) $\xrightarrow[\text{Zeff.}\downarrow\text{se ionic radii}\uparrow\text{se}]{Na^+<F^-<O^{2-}<N^{3-}}$

(ii) $\xrightarrow[\text{ionic radii}\uparrow\text{se no. of}e^-\uparrow\text{se so},\sigma\uparrow\text{se}]{Li^+<Na^+<K^+<Rb^+<Cs^+}$

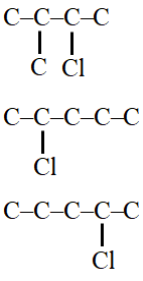
(iii) $\dfrac{N^+ < O^+}{\text{Zeff}\uparrow \text{EN}\uparrow e^-\text{s}\uparrow \text{se}\sigma\uparrow}$

(iv) $\xrightarrow[\text{Zeff}\uparrow\text{EN}\uparrow\text{EA}\uparrow]{Cl^-<K^+<Ca^{2+}<Sc^{3+}}$

57. Ans (3)

He = +48, Ne = +116, Ar = +96, Kr = +96, Xe = + 77, Rn = +68

58. Ans (1)



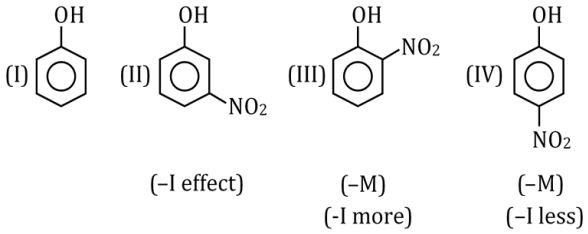
59. Ans (1)

Gauche is stable due to intramolecular H-bonding.

60. Ans (3)

Non aromatic compounds are non planar or have incomplete cyclic conjugation.

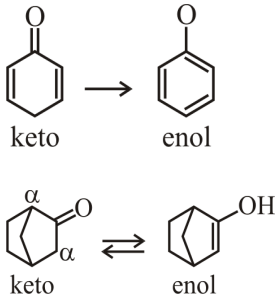
61. Ans (4)



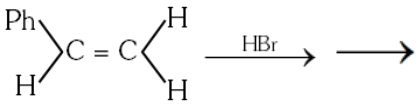
(due to H-bonding in III, it is less acidic than IV)

So, decreasing order of acidic strength ⟶ IV > III > II > I

62. Ans (3)

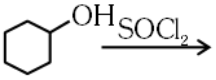


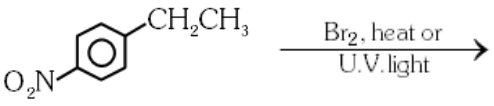
63. Ans (3)

(a) 

Electrophilic addition reaction

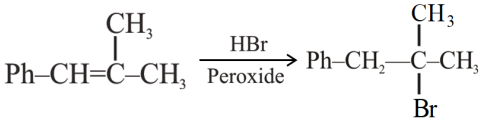
(b) $Ph-CH_2-CH=CH_2 \xrightarrow[R_2O_2]{HBr} \longrightarrow$ Free radical addition reaction

(c)  ⟶ Nucleophilic substitution reaction

(d) 

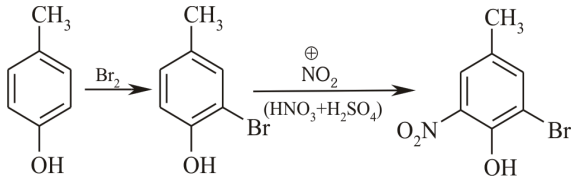
⟶ Free radical substitution reaction

64. Ans (2)



(FRAR)

65. Ans (2)



66. Ans (1)

While testing for halogens, lassaigne's extract is boiled with conc. HNO_3 to decompose Na_2S and NaCN if formed.

67. Ans (3)

Step 1 : Correct the pressure of N_2 gas

We subtract the aqueous tension (vapor pressure of water)

$P_{dry}\ N_2 = 715\ mmHg - 15\ mmHg = 700\ mmHg$

Convert mmHg at atm:

$$P = \frac{700}{760} \approx 0.9211 \ \ atm$$

Step 2 : Use Ideal Gas law to find moles of N_2

$$PV = nRT \Rightarrow n = \frac{PV}{RT}$$

Substitute :

$$n = \frac{0.9211 \times 0.050}{0.821 \times 300} \approx \frac{0.046055}{24.63} \approx 0.00187 \ \ mol$$

Step 3 : Calculate mass of nitrogen gas

Mass of N_2 = moles × molar mass of N_2 = 0.00187 × 28 ≈ 0.05236 g

Step 4 : Calculate % nitrogen

$$\%N \left(\frac{\text{mass of } N_2}{\text{mass of compound}}\right) \times 100 = \left(\frac{0.05236}{0.3}\right) \times 100 \approx 17.45\%$$

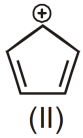
68. Ans (4)

$K_b \propto$ EDG

69. Ans (1)

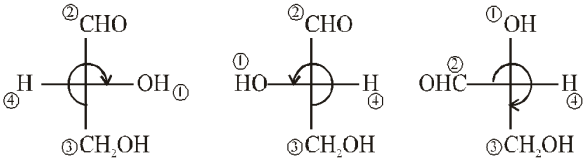


Both I and III are aromatic carbocations and hence, more stable than II which is antiaromatic.



Out of I and III; III is more stable since, it has greater number of resonance structures (7) while I has 4 resonance structures.

70. Ans (4)



[A] ® [B] Ⓢ [C] ®

(1) A and B are enantiomers

(2) A and C are identical

(3) B and C are enantiomers

(4) A and B are enantiomers

71. Ans (2)

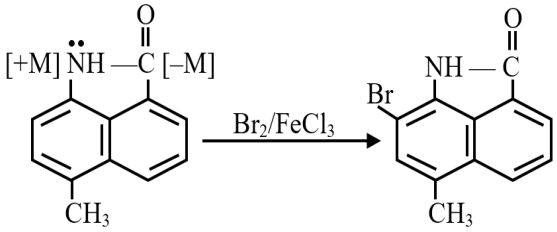
Question Explanation :

This question is asking about Product of Reaction.

Concept :

This question is based on electrophile attack at electron rich site.

Solution :



Final Answer :

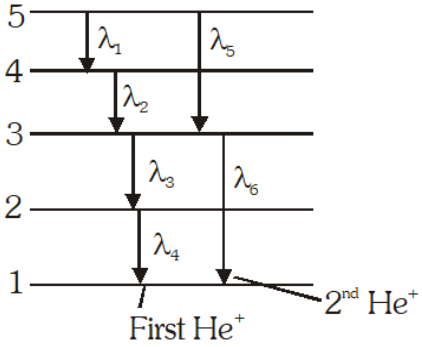
Option (2)

72. Ans (4)

A = $CH_3-\underset{\underset{O}{\|}}{C}-CH_3$, B = $CH_3-CH_2-CH_3$

C = $CH_3-\underset{\underset{Br}{|}}{CH}-CH_3$

73. Ans (3)



74. Ans (4)

$$\frac{1}{\lambda} = R\left(\frac{1}{2^2} - \frac{1}{3^2}\right) = R\left(\frac{1}{4} - \frac{1}{9}\right) = R \times \frac{5}{36}$$

$$\frac{1}{\lambda} = R\left(\frac{1}{2^2} - \frac{1}{\infty^2}\right) = \frac{R}{4}$$

$$\lambda - \lambda' = \frac{36}{5R} - \frac{4}{R} = \frac{1}{R}\left[\frac{36}{5} - 4\right] = 3.2/R_H$$

75. Ans (2)

Explain the question: Compare orbital energies for single-electron ions.

Concept: Energy depends only on shell n($E \propto -1/n^2$).

Solution:

In a single-electron species, subshells within the same principal shell are degenerate. Using the node relationships n = radial nodes (RN) + ℓ +1, we determine the shell number for each set.

S_1 is spherically symmetrical (ℓ = 0) with RN = 2, giving n = 2 + 0 + 1 = 3.

S_2 is a double dumbbell (ℓ = 2) with RN = 0, giving n = 0 + 2 + 1 = 3.

S_3 has orbital angular momentum zero (ℓ = 0) and RN = 3, giving n = 3 + 0 + 1 = 4.

S_4 has one angular node (ℓ = 1) and one radial node (RN = 1), giving n = 1 + 1 + 1 = 3.

As energy increases with n in the order n = 1 < n = 2 < n = 3 < n = 4, we find that S_1, S_2 and S_4 have equal energy (since they all belong to shell n = 3) and S_3 has a higher energy because it belongs to shell n = 4.

Final Answer: Option (2)

76. Ans (2)

$2H_2S(g) \rightleftharpoons 2H_{2(g)} + S_{2(g)}$

P_0

$P_0(1-\alpha)$ $P_0\alpha$ $\frac{P_0\alpha}{2}$

$P = P_0 + \frac{P_0\,\alpha}{2}$ $P_0 = \frac{2P}{2+\alpha}$

$$K_P = \frac{(P_0\alpha)^2\left(\frac{P_0\alpha}{2}\right)}{(P_0(1-\alpha))^2} = \frac{P_0\alpha^3}{2(1-\alpha)^2} = \frac{2P\alpha^3}{2(2+\alpha)(1-\alpha)^2}$$

$$K_P = \frac{P\alpha^3}{(2+\alpha)(1-\alpha)^2}$$

77. Ans (3)

$2NO + O_2 \rightleftharpoons 2NO_2$, K_1 ....(1)

$NO_2 + SO_2 \rightleftharpoons SO_3 + NO$ K_2 ...(2)

$2SO_3 \rightleftharpoons 2SO_2 + O_2$, K_3 ...(3)

(1) + (3)

$2NO + 2SO_3 \rightleftharpoons 2NO_2 + 2SO_2$ $K_1 \times K_3$ ....(4)

$\frac{1}{2}(4)^{-1}$

$$\boxed{\left(\frac{1}{K_1 \times K_3}\right)^{1/2} = K_2}$$

78. Ans (2)

$CaCl_2 \longrightarrow Ca^{+2} + 2Cl$

0.01 0.01 2×0.0

$Ca(OH)_2 \rightleftharpoons Ca^{+2} + 2OH^{\Theta}$

S 2S

(S' + 0.01) 2S' (in $CaCl_2$)

$k_{sp} = [Ca^{+2}]\,[OH^-]^2 = (S' + 0.01)\,(2s')^2$

$= (S' + 0.01) \times 4s'^2$

$k_{sp} = 0.01 \times 4.S'^2$ $S' = \left(\frac{k_{sp}}{4 \times 0.01}\right)^{1/2}$

$S' = \left(\frac{4 \times 10^{-6}}{4 \times 10^{-2}}\right)^{1/2} = 10^{-2}$

$[OH^-] = 2 \times 10^{-2}$ $p^{OH} = 2 - \log 2$

$p^{17} = 14 - (2 - \log 2) = 12 + \log 2$

79. Ans (1)

$\Delta G_{sys.} = -T\Delta S_{total}$

$= -(400)(-40)$

$= 16000$ kJ mol^{-1}

$\Delta G_{sys.} = \Delta H_{sys.} - T\Delta S_{sys.}$

$16000 = 2000 - (400)\,\Delta S_{sys.}$

$\Delta S_{sys.} = -35$ kJ mol^{-1} K^{-1}

80. Ans (3)

$\Delta G = \Delta H - T\Delta S$

For spontaneous process, ΔG is −ve i.e., < 0

$\therefore$ $\Delta H - T\Delta S < 0$ or $T\Delta S > \Delta H$

$\therefore$ $T\Delta S$ and ΔH are +ve

Thus $\Delta G = \Delta H - T\Delta S$ become −ve.

81. Ans (1)

at equilibrium

$\Delta H = T \cdot \Delta S$

$H_2O(S) \longrightarrow H_2O(\ell)$

$\Delta S = S_{H_2O(\ell)} - S_{H_2O(S)} = 60 - 38.2 = 21.8$ J/mol−k

$\Delta H = 273 \times 21.8$

$= 5951.4$ J/mol

82. Ans (4)

$n_{H_2} = \frac{64}{2} = 32$ mol

$n_{O_2} = \frac{64}{32} = 2$ mol

1 mol of O_2 requires = 2 mol of H_2

2 mol of O_2 requires = 4 mol of H_2

Since H_2 is present in excess, therefore, O_2 is the limiting reagent

(1) Wrong

(2) So, O_2 is the limiting reagent

(3) 2mol of O_2 = 4 moles of $H_2O = 4 \times 18$

= 72 g H_2O

Moles of H_2 left = 32 − 4 = 28 mol

$= 28 \times 2 = 56$ g H_2

Mixture contains = (72 g H_2O + 56 g H_2)

83. Ans (2)

According to Le-chatelier's principle

84. Ans (3)

$H_2O(s) \rightleftharpoons H_2O(l)$; ΔH = +ve

Volume of ice > Volume of water

85. Ans (2)

$A_2O_x + MnO_4^{\Theta} \rightarrow AO_3^{\Theta} + Mn^{+2}$

Equivalent of A_2O_x = Equivalent of $KMnO_4$

$1.5 \times 10^{-3} \times (10-2x) = 0.03 \times 5 \times \frac{40}{1000}$

x = 3

so the formula of the compound A_2O_3

86. Ans (4)

Number of atoms of iodine on reactant side = number of atoms of Iodine on product side

$2 + x = 6 \times 2$

X = 10

$2IO_3^- + 10I^- + 12H^+ \rightarrow 6I_2 + 6H_2O$

87. Ans (3)

$$H_2O(\ell) \longrightarrow H_2(g) + \frac{1}{2}O_2(g)$$

moles $\frac{144}{18} = 8$

after reaction — 8 — 4

$w = -\Delta n_g RT$

$= -(8 + 4) \times 8.3 \times 300$ J

88. Ans (2)

$nC_2H_4(g) \longrightarrow (C_2H_4)_n(g)$; $\Delta H = -6000$ kJ

$-6000 = (n \times 540) - (2n \times 330) \Rightarrow n = 50$

89. Ans (1)

Explanation: match list-1 with list-2.

Concept: calculation of mole

Solution:

(P) 0.5 mole of SO_2(g)

- Volume: At STP (1 atm, 273 K), 1 mole occupies 22.4 L . So, 0.5 mol = 11. 2L → Matches (1).

- Atoms: SO_2 has 3 atoms per molecule. Total atoms = $0.5 \times 3 \times N_A = 1.5 \times N_A$ → Matches (2).

- Electrons: S(16) + 2 × O(8) = 32 electrons per molecule. Total = $0.5 \times 32 \times N_A = 16N_A$ electrons.

- Protons: Same as electrons for neutral molecules = $16N_A$ → Matches (5).

(Q) 1 gm molecule of O_2(g)

Note: "1 gm molecule" is a traditional term for 1 mole.

- Mass: Molar mass of O_2 is 32g/mol. So, 1 mole = 32 g → Matches (4).

- Electrons/Protons: 2 × O(8) = 16 per molecule. Total = $1 \times 16 \times N_A = 16N_A$ → Matches (5).

(R) 44 gm of CO_2 gas

- Moles: Molar mass of CO_2 = 12 + (2 × 16) = 44 g/mol. So, 44 g = 1 mole.

- Electrons: C(6) + 2 × O(8) = 22 electrons per molecule. Total = $1 \times 22 \times N_A = 22N_A$ → Matches (3).

(S) 0.5 mole of O_3(g)

- Volume: 0.5 mole ×22.4 L/mol = 11. 2L → Matches (1).

- Atoms: O_3 has 3 atoms per molecule. Total = $0.5 \times 3 \times N_A = 1.5 \times N_A$ → Matches (2).

- Mass: Molar mass = 3 × 16 = 48 g/mol. Mass = 0.5 × 48 = 24 g.

P → 1, 2, 5; Q → 4, 5; R → 3; S → 1, 2

Final answer: option (1)

90. Ans (3)

(P) $NH_4OH + HCl \longrightarrow NH_4Cl + H_2O$

0.1 — 0.1 — — —

0 — 0 — 0.1m

$(H^+) = \sqrt{\frac{K_W \times C}{K_b}} = \sqrt{\frac{10^{-14}}{10^{-5}} \times 10^{-1}} = 10^{-5}$

(Q) $CH_3COONa + HCl \longrightarrow CH_3COOH + NaCl$

0.1 — 0.1 — — —

0 — 0 — 0.1M

$[H^+] = \sqrt{K_a . C} = \sqrt{10^{-5} \times 10^{-1}} = 10^{-3}$

pH = 3

(R) $H_2CO_3 + NaOH \longrightarrow NaHCO_3 + H_2O$

A — a/2 — —

a/2 — 0 — a/2

pH = pka = 6

(S) $Mg(OH)_2(s) \rightleftharpoons Mg^{2+}(aq.) + 2OH^-(aq.)$

(s) — (s) — 2s

$Ksp = 4s^3$

$5 \times 10^{-16} = 4s^3$

$s^3 = \frac{5}{4} \times 10^{-16} = (1.25 \times 10^{-16})$

$= (125 \times 10^{-18})^{1/3}$

$= 5 \times 10^{-6}$

$(OH^-) = 2s = 10^{-5}$

pOH = 5

pH = 9

91. Ans (2)

NCERT-XI_Pg # 4,5

92. Ans (1)

NCERT-XI_Pg # 7

93. Ans (3)

NCERT-XI_Pg # 11

94. Ans (3)

NCERT-XI_Pg # 13

95. Ans (2)

NCERT-XI_Pg # 15

96. Ans (4)

NCERT Pg. # 15

97. Ans (3)

NCERT 11th Pg. # 17,18

98. **Ans (4)**
NCERT Pg. # 20,21

99. **Ans (4)**
NCERT Pg. # 24

100. **Ans (2)**
NCERT Pg. # 29

101. **Ans (2)**
NCERT Page No. 32

102. **Ans (4)**
NCERT-XI, Pg. # 29, 30

103. **Ans (4)**
NCERT (XI) Pg # 30-32

104. **Ans (2)**
NCERT, Pg. # 30,32

105. **Ans (4)**
NCERT Pg. # 33

106. **Ans (2)**
NCERT-XI, Pg # 96

107. **Ans (3)**
NCERT Pg. # 88

108. **Ans (2)**
NCERT-XI, Pg. #127

109. **Ans (3)**
NCERT Pg. #127,128

110. **Ans (3)**
Question Explanation: We are asked to match DNA amounts in different cell types/stages when a human sperm cell has 14 pg DNA (haploid).
Concept: Cell division
Solution: In humans:
• A haploid (n; 1C) sperm has 14 pg DNA.
• A diploid somatic cell at G_0/G_1 (2n; 2C) therefore has 28 pg DNA.
• After DNA replication in S phase (4C), a primary spermatocyte or somatic cell in G_2 has 56 pg DNA.
• At metaphase II in meiosis, the cell is haploid (n) but has duplicated chromosomes (2C) ⟶ 28 pg DNA.
So, only the statement which states the amount of DNA in a somatic cell at G_0 (28 pg) matches correctly with the given sperm DNA content.
Final answer: Option 3.

111. **Ans (1)**
NCERT Pg. # 128

112. **Ans (3)**
NCERT XI Page No. # 126

113. **Ans (4)**
NCERT-XI_Pg # 93

114. **Ans (2)**
NCERT-XI_Pg # 99

115. **Ans (2)**
NCERT-XI_Pg # 101

116. **Ans (1)**
NCERT-XI_Pg # 126, 127

117. **Ans (2)**
Question Explanation:
This question asks you to identify the types of glycosidic bonds (A and B) present in glycogen, based on the diagram.
Concept- Biomolecules ⟶ Polysaccharides: glycogen structure
Explanation
- Bond A is at the branch point, so it must be an $\alpha-1,6$ glycosidic bond.
- Bond B is in the straight chain, so it must be an $\alpha-1,4$ glycosidic bond.

This matches option 2.
Final Answer: Option 2

118. **Ans (2)**
NCERT XI, Pg # 146

119. **Ans (2)**
NCERT Pg. # 158

120. **Ans (2)**
NCERT-XII, Pg. # 158

121. **Ans (3)**
NCERT-XI_Pg # 149

122. **Ans (2)**
NCERT-XI Pg # 212
Statement (i), (iii), (iv) are not related

123. Ans (4)
NCERT-XI Pg#216

124. Ans (4)
NCERT-XI Eng. Pg. # 217 (Topic -Reduction & Regeneration)

125. Ans (1)
NCERT XI, Pg # 211 to 212

126. Ans (2)
NCERT-XI_Pg # 150

127. Ans (1)
NCERT-XI, Eng. Pg. # 233

128. Ans (2)
NCERT XI Pg. No. # 160

129. Ans (4)
NCERT-XI, Pg. # 156

130. Ans (2)
NCERT-XI_Pg. # 158

131. Ans (3)
Explain Question: The question is asking : When the Respiratory Quotient (RQ) is less than 1, which statement cannot be concluded ?
Concept: Respiratory Quotient (RQ) **Solution:** When RQ < 1, it means oxygen consumption is more than carbon dioxide release. This generally occurs when fats or proteins are used as respiratory substrates.

1. Respiratory substrate could be fat . True (Fat has RQ ≈ 0.7)
2. Respiratory substrate could be protein . True (Protein has RQ ≈ 0.9)
3. In respiratory substrates oxygen amount is more than carbon. Not true because If a substrate already contains a high amount of oxygen (like organic acids), it requires less external oxygen to break down, resulting in an RQ greater than 1. When the substrate is oxygen-poor (like fats), it requires more external oxygen, leading to an RQ less than 1.
4. Might be a starved cell. True (In starvation, fats are used, giving RQ < 1)

Final answer: Option 3

132. Ans (2)
NCERT-XI_Pg # 175, 176, 177

133. Ans (4)
NCERT-XI_Pg # 173

134. Ans (4)
NCERT-XI_Page # 175

135. Ans (3)
NCERT-XI_Pg # 176

136. Ans (3)
NCERT (XIth) Pg. # 54

137. Ans (4)
NCERT XI Pg.# 51, figure 4.9(a)

138. Ans (3)
NCERT Pg.#52

139. Ans (2)
XI NCERT Pg. No :- 55

140. Ans (1)
(Old NCERT)-XI Pg.#53

141. Ans (4)
NCERT-XI, Pg. # 57,59,60, Para-4.2.11.3, 4.2.11.6, 4.2.11.7

142. Ans (4)
NCERT Pg.#42-45.

143. Ans (4)
XI NCERT Pg. No :- 101

144. Ans (4)
NCERT Page. No : 101

145. Ans (1)
NCERT-XI, Pg. # 113

146. Ans (4)
XI NCERT Pg. No :- 111, 113

147. Ans (4)
XI NCERT Pg. No :- 101, 102, 138, 139

148. Ans (1)
NCERT (XIth) Pg. # 118

149. Ans (3)
NCERT Pg # 118

150. Ans (1)
NCERT Pg # 83

151. Ans (2)
NCERT Pg.#194.

152. Ans (3)
NCERT Pg.#194, 195, 199.

153. Ans (1)
NCERT Pg.#201.

154. Ans (3)
NCERT Pg. # 194

155. Ans (1)
NCERT Pg. # 196

156. Ans (4)
NCERT Pg # 202

157. Ans (1)
NCERT Pg # 190

158. Ans (1)
NCERT Pg. # 269

159. Ans (2)
NCERT Pg. # 270

160. Ans (2)
NCERT Pg # 275

161. Ans (3)
NCERT Pg. # 270, 271

162. Ans (2)
NCERT Pg. # 189

163. Ans (4)
NCERT, Pg. # 212

164. Ans (2)
NCERT Pg. No. # 207; Fig. 16.3

165. Ans (1)
NCERT Pg # 208

166. Ans (2)
NCERT Pg # 209

167. Ans (3)
NCERT, Pg. # 212

168. Ans (2)
NCERT Pg.#230,231,242,243.

169. Ans (1)
NCERT Pg. No. # 230

170. Ans (1)
NCERT Pg # 233

171. Ans (4)
NCERT XI Pg. # 236

172. Ans (3)
NCERT Pg. # 241, 243, 244, 245

173. Ans (2)
NCERT Pg.# XI- 240,241,242,244.

174. Ans (1)
NCERT Pg. 245

175. Ans (1)
NCERT Pg. # 241

176. Ans (4)
NCERT Pg # 219

177. Ans (3)
NCERT Pg # 218

178. Ans (2)
NCERT Page No. # 225

179. Ans (1)
NCERT Pg.#227.

180. Ans (4)
NCERT Pg.#218.